LittleHeart

AI heart murmur screening for children · UnivaBio 2026: AI for Human Health

The problem

About 1 in 100 babies is born with a heart defect, and rheumatic heart disease affects tens of millions of people, most of them young and living in low- and middle-income countries. Both are often first noticed as a heart murmur. But recognising a murmur through a stethoscope takes years of training, and the echocardiogram that confirms it, and the cardiologists who read it, are scarce exactly where these diseases are most common. Many children are diagnosed only when the heart is already damaged.

Our solution

LittleHeart is a web app that lets a community health worker, school nurse or GP record a child's heart at the four standard valve areas and instantly learn whether a murmur is likely: *refer for echocardiogram*, *no murmur detected*, or *recording unclear, record again*. It turns an expert skill into a consistent second opinion so scarce echocardiograms go to the children who need them.

Key features

• **Three ways in**: real held-out patient cases, file upload, or live microphone/stethoscope recording.
• **Per-valve results** on an interactive chest diagram.
• **Explainable**: spectrogram of each recording with a moving playhead and a murmur-likelihood timeline.
• **Private**: the AI runs in the browser (ONNX Runtime Web); no audio is uploaded.
• **Transparent validation tab** and an honest "working / not yet / limits" table.
• Education tab on murmurs, recording technique and next steps; phone layout and dark mode.

How the AI works

Trained on the CirCor DigiScope dataset (PhysioNet): 3,163 recordings from 942 children in Brazil, labelled by cardiologists. Each recording becomes a log-mel spectrogram (25-1000 Hz), cut into 4-second windows. A small convolutional neural network (about 61,000 weights, PyTorch) scores each window as murmur / no murmur / unclear; scores are averaged per valve, and a child is referred if any valve crosses a tuned threshold. A **heartbeat safety check** first refuses recordings with no 40-220 bpm rhythm (noise, tones, most speech), so the AI never guesses on sounds it was not trained for. The same code exists in Python and JavaScript and automated tests check they match.

Results

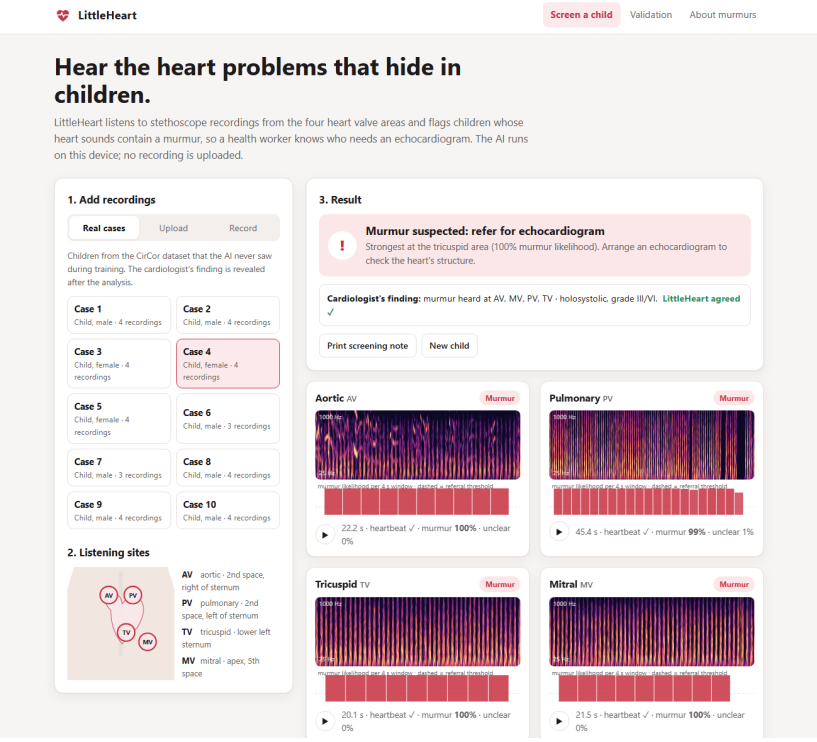
5-fold cross-validation, split by child, on 932 children: official PhysioNet 2022 murmur score **0.78 ± 0.05** (simple baseline 0.60; for context, the best challenge team scored 0.78 on a harder hidden test set, so this is not a like-for-like comparison). It caught **87%** of children with murmurs and cleared **91%** of those without (AUC 0.94). 10 demo children were never used in training.

Honest limits

A screening aid, not a diagnosis or medical device. It still misses some murmurs, was trained on one population and one stethoscope, cannot yet tell innocent from pathological murmurs, and has not been tested prospectively or with phone microphones. All of this is stated in the app.

Next steps

Test with low-cost stethoscope adapters, validate on other datasets and populations, add murmur timing and grading, multilingual interface, and a pilot with a school RHD-screening programme.



Tech: Python, PyTorch, scikit-learn, NumPy (training) · HTML, CSS, JavaScript, ONNX Runtime Web (app) · Data: CirCor DigiScope Phonocardiogram Dataset, Oliveira et al. 2022, PhysioNet, CC BY 4.0.